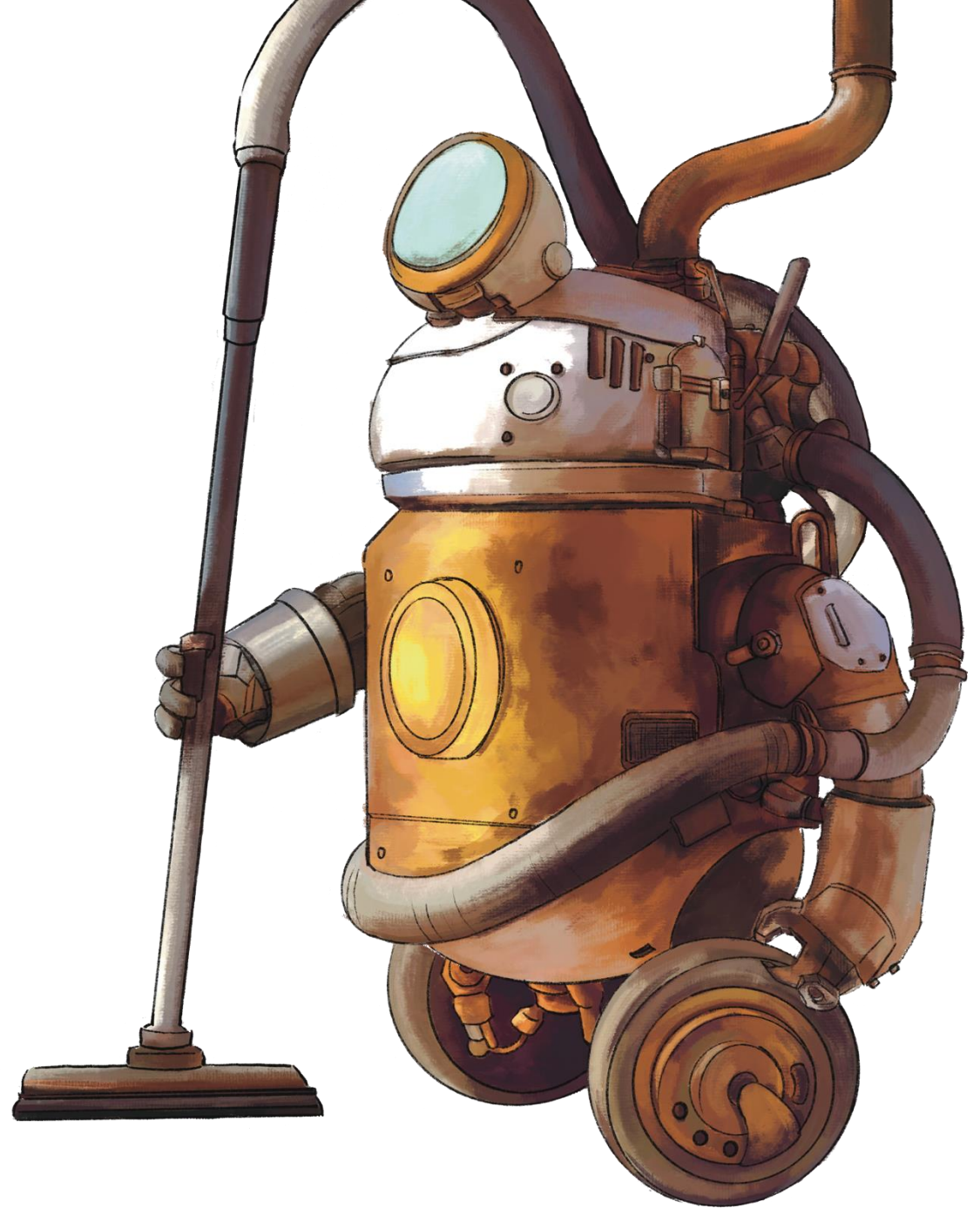


CS 3630, Fall 2025

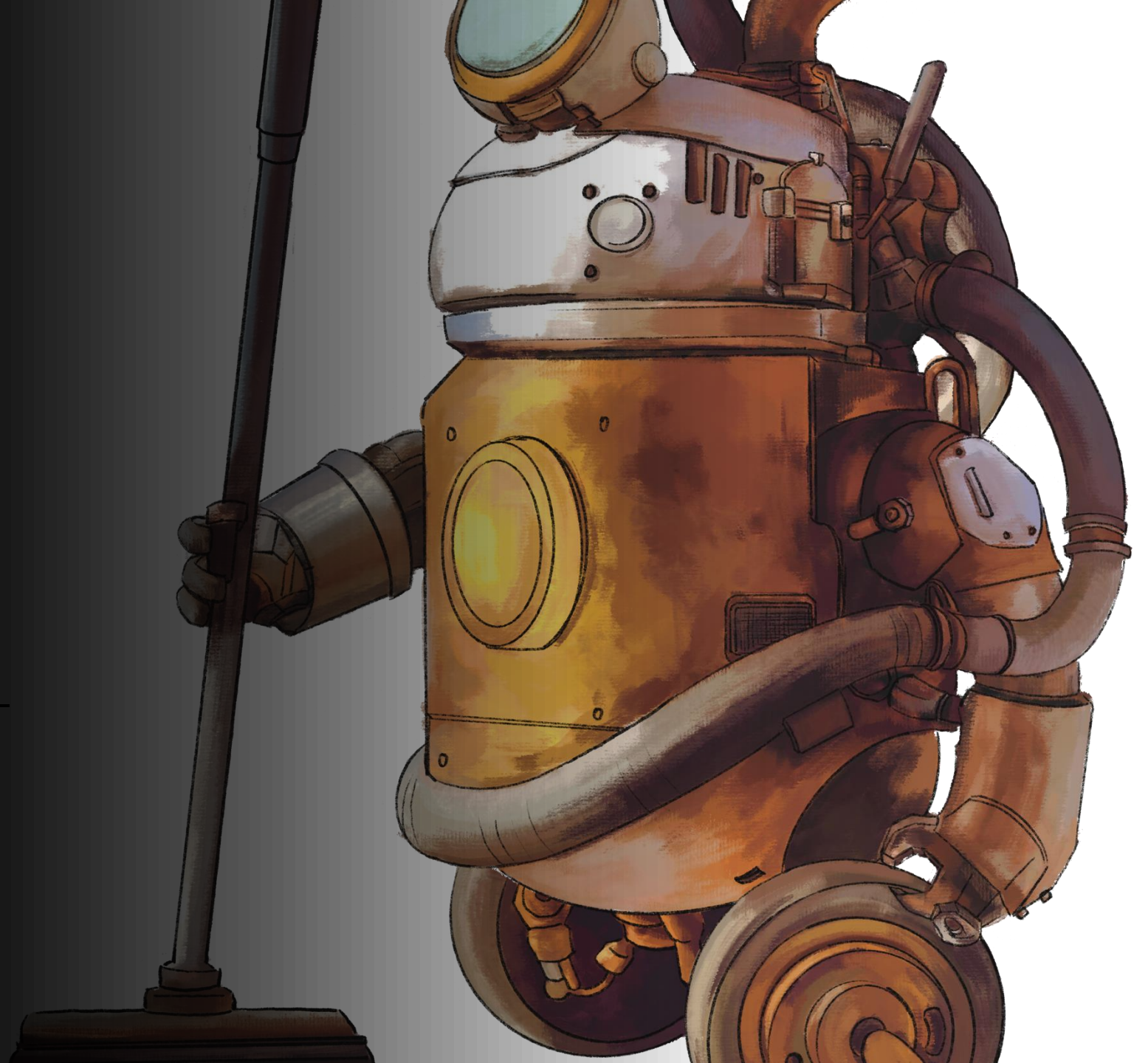
Lecture 7:

A Vacuum Cleaning Robot: Actions & Markov Chains





Lecture 6 Recap



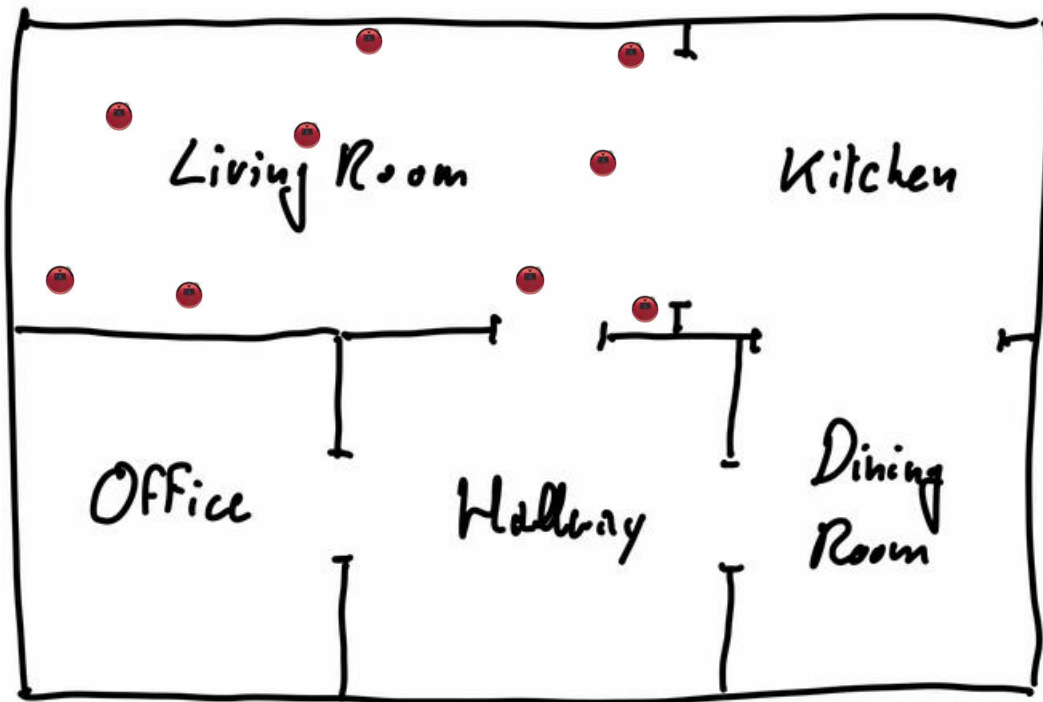
State Space



A typical vacuum cleaning robot.

For this robot, the state, X , is defined as the room in which the robot is currently located:

$$X \in \{\text{living room, kitchen, office, hallway, dining room}\}$$



For all of the robot locations shown here, we have:

$$X = \text{living room}$$

The exact location within the living room is not relevant for this robot.

To simplify notation, we'll sometimes write $X \in \{L, K, O, H, D\}$.

Discrete time systems



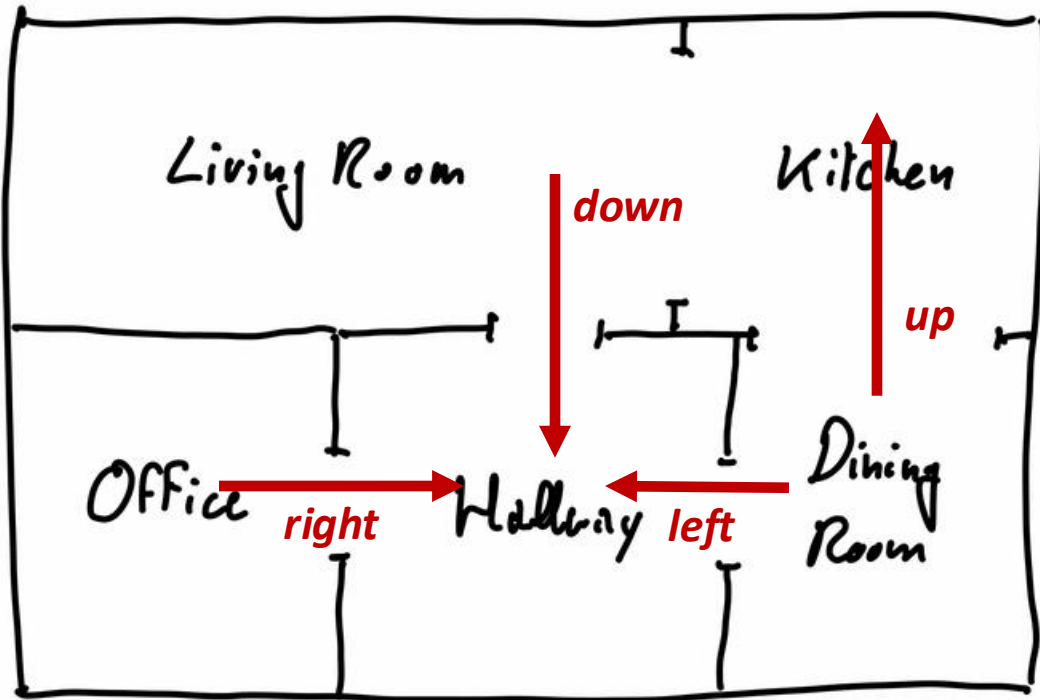
- For our trash sorting robot, there was no need to consider the passing of time.
- For our vacuum cleaning robot, the passing of **time is important**.
- We only need to keep track of state at discrete time instants, $t \in \{t_0, t_1 \dots\}$.
- **We represent state at time t by X_t , and we'll simplify by using $t \in \{0, 1, 2 \dots\}$.**
- The initial state of the robot (i.e., when it wakes up) is therefore: $X_0 = office$.

Actions

Our robot has four actions:

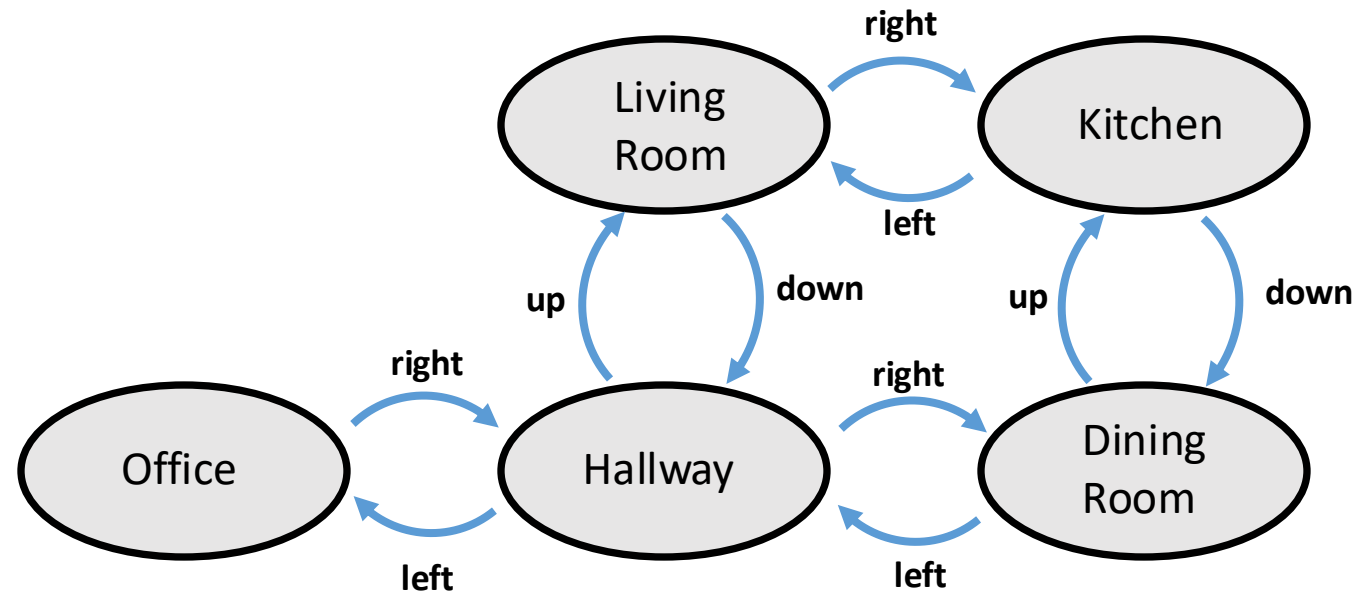
up, down, left, right.

- Effects of actions are context dependent.
- Actions potentially cause a change in state.
- Executing an action in state X_t produces state X_{t+1}



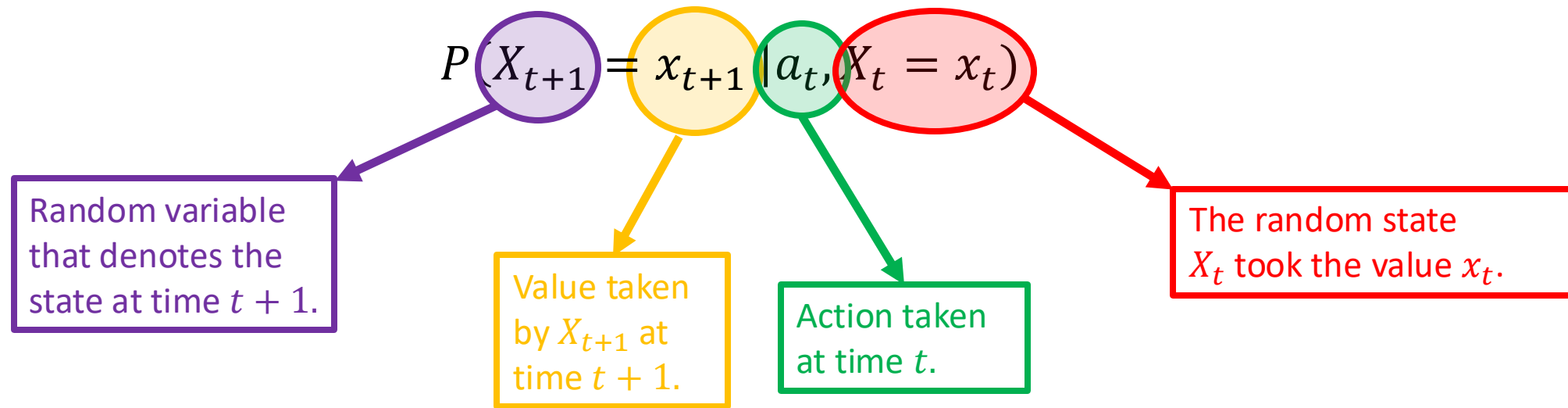
We can represent this by a slight modification to our state space:

- Instead of using an undirected graph, use a directed graph.
- Each edge (u, v) corresponds to an action meant to change the state from $x_t = u$ to $x_{t+1} = v$.
- Sadly, our actions are not deterministic, so we need to do a bit more work.



Uncertainty in the effects of actions

- We will model uncertainty in the effects of actions by using conditional probability distributions.
- In particular, we define the conditional probability distribution for the next state, X_{t+1} , given that the current state, X_t is room x_t , and that action a_t was executed at time t .



Example: If we are in the *Office* at time t and execute the *move right* action, $P(X_{t+1} = H | right, X_t = O)$ denotes the conditional probability of arriving to the *Hallway*.

The Markov property

- Using our Markov property, we can write

$$P(X_3 = x_3 \mid \boxed{right, up, right}, \boxed{X_0 = O, X_1 = H}, X_2 = L) = P(X_3 = x_3 \mid \boxed{right}, \boxed{X_2 = L})$$

The diagram illustrates the Markov property by showing how the probability of a future state depends only on the current state. Four boxes below the equation provide context for the terms in the probability expression:

- A blue box labeled "What the robot has done before time t ." has an arrow pointing to the sequence of actions $right, up, right$ in the first part of the equation.
- An orange box labeled "Where the robot has been before time t ." has an arrow pointing to the sequence of previous states $X_0 = O, X_1 = H$ in the first part of the equation.
- A green box labeled "What the robot does now, at time t ." has an arrow pointing to the action $right$ in the second part of the equation.
- A purple box labeled "Where the robot is now, at time t ." has an arrow pointing to the current state $X_2 = L$ in the second part of the equation.

Our Markov assumption:

$$P(X_{t+1} = x_{t+1} \mid a_0, \dots, a_t, X_0 = x_0, \dots, X_t = x_t) = P(X_{t+1} = x_{t+1} \mid a_t, X_t = x_t)$$

Actions: Part 2

- Our vacuum cleaning robot has four actions:
 - Move *left, right, up, or down* (relative to the map of the house)
- Effects of actions are probabilistic.
- Effects of actions depend on the current state.
 - ***Use conditional probabilities to model the effects of actions.***
- For a specific sequence of actions (e.g., *up, right, down, left*) computing probabilities for states in the distant future seems complicated.
 - ***Happily, thanks to the Markov property, these computations are not so difficult.***

Conditional probability distributions for actions

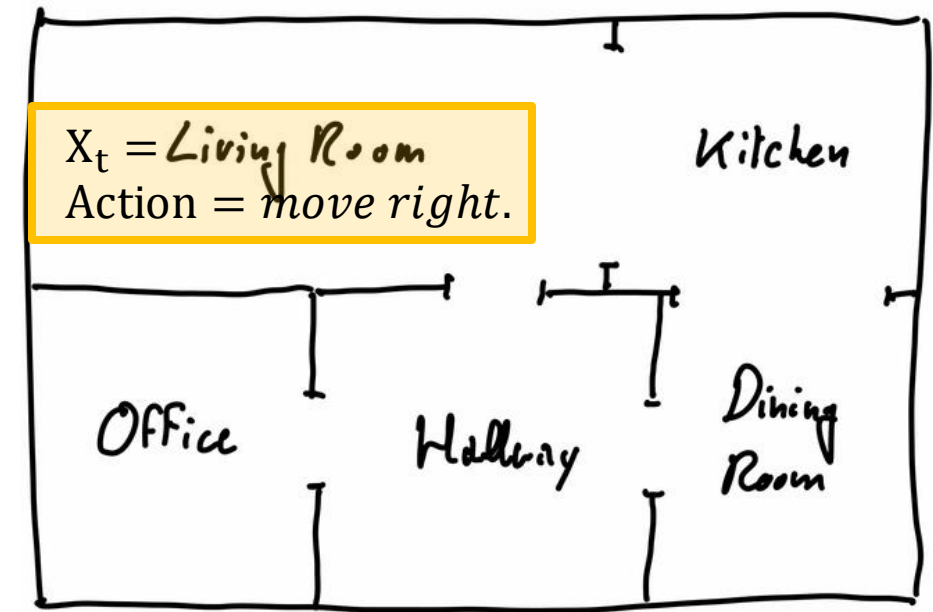
- Thanks to our Markov assumption, all necessary knowledge about the probabilistic effects of actions is included in our conditional probability tables.
- For example, if $X_t = L$, we can write conditional probability distributions for each of the four possible actions.
- In our example scenario, a reasonable distribution could be:

X_t	a_t	Living Room	Kitchen	Office	Hallway	Dining Room
Living Room	L	1	0	0	0	0
Living Room	R	0.2	0.8	0	0	0
Living Room	U	1	0	0	0	0
Living Room	D	0.2	0	0	0.8	0

Conditional probability distributions for actions

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Living Room	U	1	0	0	0	0
Living Room	D	0.2	0	0	0.8	0



- *Regardless of how we came to be in the Living Room, if we now execute the action move right, we arrive to the Kitchen with probability 0.8, and stay in the Living Room with probability 0.2.*

Conditional probability distributions for actions

- Thanks to our Markov assumption, we can encapsulate all necessary knowledge about the probabilistic effects of actions using conditional probability tables.
- For example, if $X_t = L$, we can write conditional probability distributions for each of the four possible actions.
- In our example scenario, a reasonable distribution could be:

		X_{t+1}				
X_t	a_t	Living Room	Kitchen	Office	Hallway	Dining Room
Living Room	L	1	0	0	0	0
Living Room	R	0.2	0.8	0	0	0
Living Room	U	1	0	0	0	0
Living Room	D	0.2	0	0	0.8	0

Taken together, these four rows give the Conditional Probability Table for:

- Arriving to the each of the five possible rooms for X_{t+1}
- Given that
 $X_t = \text{Living Room}$
- For each possible action a_t
 - Left, Right, Up, Down

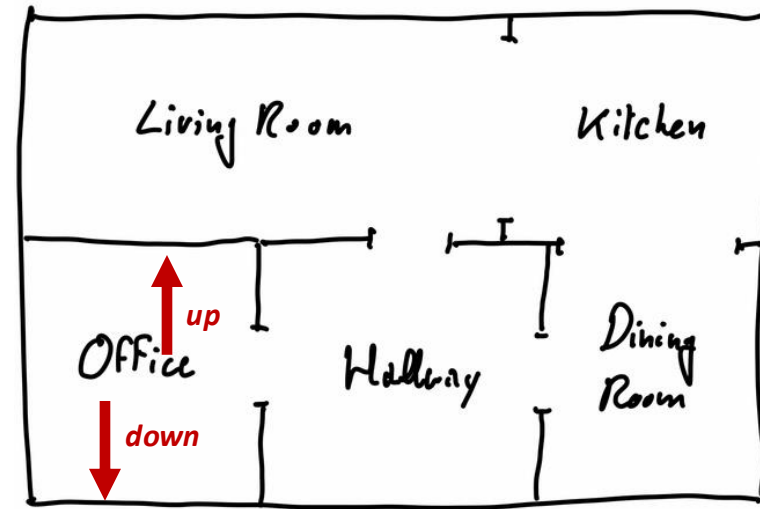
$$P(X_{t+1} = x_{t+1} | a_t, X_t = \text{Living Room})$$

Conditional probability tables

In the book, you'll find the CPTs for all rooms collected into a very large table.

X1	A1	Living Room	Kitchen	Office	Hallway	Dining Room
Living Room	L	1	0	0	0	0
Living Room	R	0.2	0.8	0	0	0
Living Room	U	1	0	0	0	0
Living Room	D	0.2	0	0	0.8	0
Kitchen	L	0.8	0.2	0	0	0
Kitchen	R	0	1	0	0	0
Kitchen	U	0	1	0	0	0
Kitchen	D	0.2	0	0	0	0.8
Office	L	0	0	1	0	0
Office	R	0	0	0.2	0.8	0
Office	U	0	0	1	0	0
Office	D	0	0	1	0	0
Hallway	L	0	0	0.8	0.2	0
Hallway	R	0	0	0	0.2	0.8
Hallway	U	0.8	0	0	0.2	0
Hallway	D	0	0	0	1	0
Dining Room	L	0	0	0	0.8	0.2
Dining Room	R	0	0	0	0	1
Dining Room	U	0	0.8	0	0	0.2
Dining Room	D	0	0	0	0	1

This table was constructed by hand, with intuitively reasonable probability values.



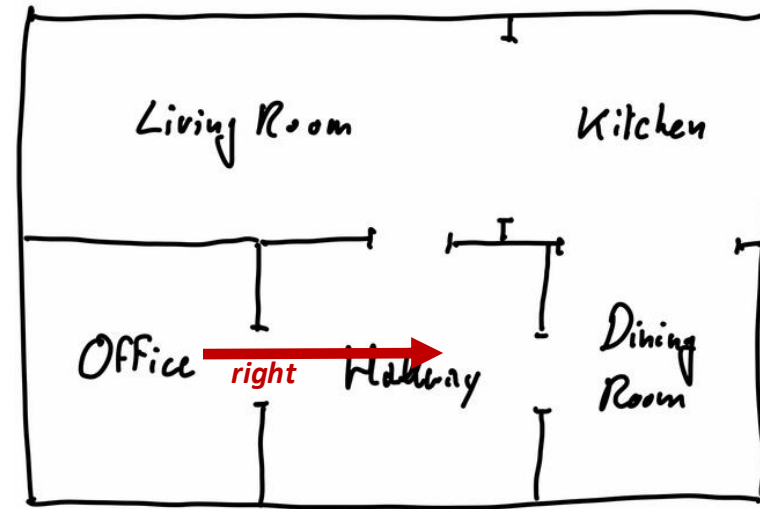
If the robot is in the **Office**, then moving **up** or moving **down** will not allow the robot to change rooms.

Conditional probability tables

In the book, you'll find the CPTs for the four actions collected into a very large table.

X1	A1	Living Room	Kitchen	Office	Hallway	Dining Room
Living Room	L	1	0	0	0	0
Living Room	R	0.2	0.8	0	0	0
Living Room	U	1	0	0	0	0
Living Room	D	0.2	0	0	0.8	0
Kitchen	L	0.8	0.2	0	0	0
Kitchen	R	0	1	0	0	0
Kitchen	U	0	1	0	0	0
Kitchen	D	0.2	0	0	0	0.8
Office	L	0	0	1	0	0
Office	R	0	0	0.2	0.8	0
Office	U	0	0	1	0	0
Office	D	0	0	1	0	0
Hallway	L	0	0	0.8	0.2	0
Hallway	R	0	0	0	0.2	0.8
Hallway	U	0.8	0	0	0.2	0
Hallway	D	0	0	0	1	0
Dining Room	L	0	0	0	0.8	0.2
Dining Room	R	0	0	0	0	1
Dining Room	U	0	0.8	0	0	0.2
Dining Room	D	0	0	0	0	1

This table was constructed by hand, with intuitively reasonable probability values.



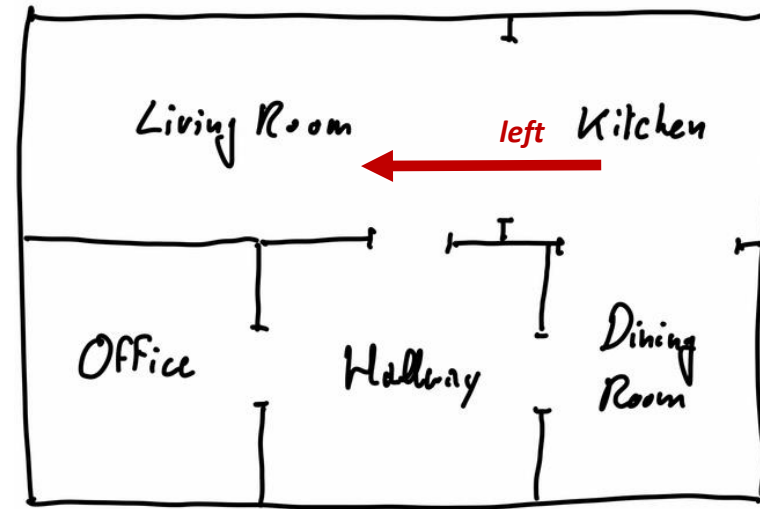
If the robot is in the **Office** and moves **right**, it will stay in the **Office** (prob = 0.2) or arrive to the **Hallway** (prob = 0.8)

Conditional probability tables

In the book, you'll find the CPTs for the four actions collected into a very large table.

X1	A1	Living Room	Kitchen	Office	Hallway	Dining Room
Living Room	L	1	0	0	0	0
Living Room	R	0.2	0.8	0	0	0
Living Room	U	1	0	0	0	0
Living Room	D	0.2	0	0	0.8	0
Kitchen	L	0.8	0.2	0	0	0
Kitchen	R	0	1	0	0	0
Kitchen	U	0	1	0	0	0
Kitchen	D	0.2	0	0	0	0.8
Office	L	0	0	1	0	0
Office	R	0	0	0.2	0.8	0
Office	U	0	0	1	0	0
Office	D	0	0	1	0	0
Hallway	L	0	0	0.8	0.2	0
Hallway	R	0	0	0	0.2	0.8
Hallway	U	0.8	0	0	0.2	0
Hallway	D	0	0	0	1	0
Dining Room	L	0	0	0	0.8	0.2
Dining Room	R	0	0	0	0	1
Dining Room	U	0	0.8	0	0	0.2
Dining Room	D	0	0	0	0	1

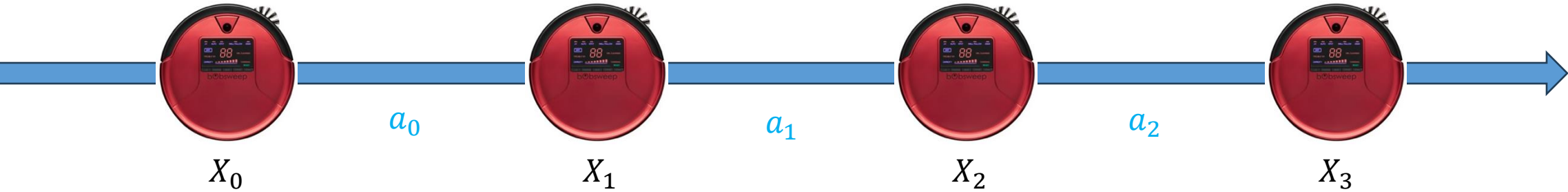
This table was constructed by hand, with intuitively reasonable probability values.



If the robot is in the **Kitchen** and moves **left**, it will stay in the **Kitchen** (prob = 0.2) or arrive to the **Living Room** (prob = 0.8)

Posterior probabilities

$$P(X_3 = x_3 \mid a_0, a_1, a_2, X_0 = x_0)$$



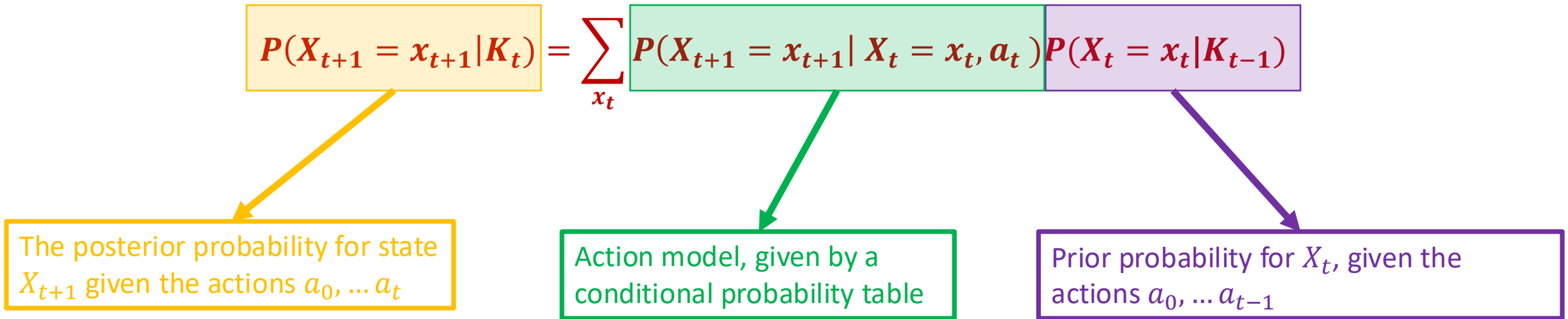
- Suppose we start in the Office, and execute a sequence of commands $a_0, \dots a_t$.
- What should we believe about the state of the robot at time $t + 1$?
- For each possible value, x_{t+1} , that can be assigned to X_{t+1} we want to determine:

$$P(X_{t+1} = x_{t+1} \mid a_0, \dots a_t, X_0 = x_0)$$

- How can we compute this?
- Is it necessary to do a long chain of reasoning all the way back to $t = 0$ every time we execute an action?

The *goal* is our location (PMF) given all actions so far:

- The result we will arrive at:



Posterior probabilities

- The law of total probability states:

$$P(A) = \sum P(A|B_t)P(B_t)$$

- We can condition everything on some context, K_t , to obtain:

$$P(A | K_t) = \sum P(A | B_t, K_t)P(B_t | K_t)$$

- Define the context at time t to be $K_t \triangleq a_0, \dots, a_t = K_{t-1}, a_t$.
- Let B_t be the event $B_t \triangleq X_t = x_t$.
- Let A be the event $A \triangleq X_{t+1} = x_{t+1}$.
- Then we can write

$$P(X_{t+1} = x_{t+1} | K_t) = \sum_{x_t} P(X_{t+1} = x_{t+1} | X_t = x_t, K_t)P(X_t = x_t | K_t)$$

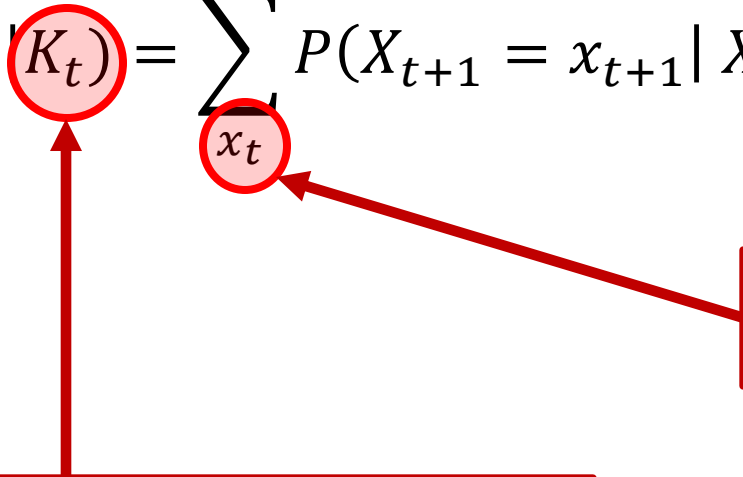
- Or: we must come from somewhere!

Posterior probabilities

- We can rewrite

$$P(X_{t+1} = x_{t+1} \mid K_t) = \sum P(X_{t+1} = x_{t+1} \mid X_t = x_t, K_t) P(X_t = x_t \mid K_t)$$

using the fact that $K_t = K_{t-1}, a_t$:

$$P(X_{t+1} = x_{t+1} \mid K_t) = \sum_{x_t} P(X_{t+1} = x_{t+1} \mid X_t = x_t, K_{t-1}, a_t) P(X_t = x_t \mid K_{t-1}, a_t)$$


Sum over all possible values of X_t :

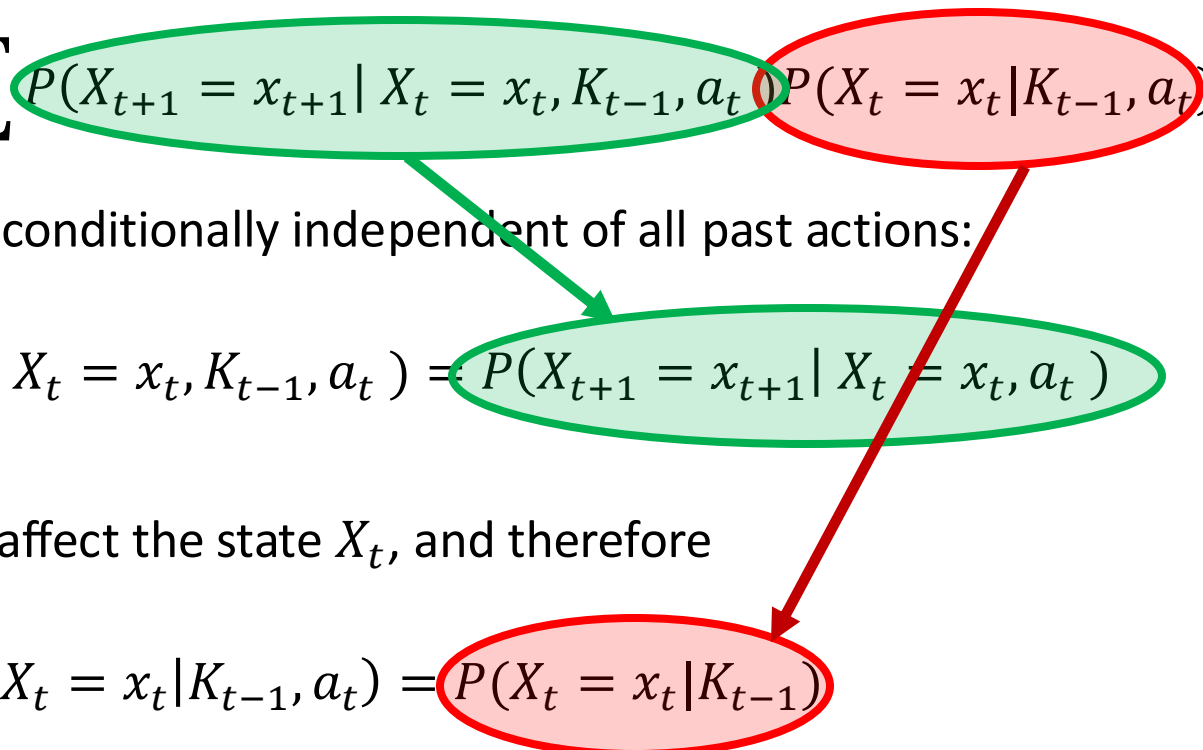
➤ Living Room, Kitchen, Office, Hallway, Dining Room

Complete history of the robot's actions to date:

➤ $K_t = a_0, \dots, a_t$

Posterior probabilities

- We can now apply our Markov assumption to the equation

$$P(X_{t+1} = x_{t+1} | K_t) = \sum_{x_t} P(X_{t+1} = x_{t+1} | X_t = x_t, K_{t-1}, a_t) P(X_t = x_t | K_{t-1}, a_t)$$


- Given $X_t = x_t$, the next state X_{t+1} is conditionally independent of all past actions:

$$P(X_{t+1} = x_{t+1} | X_t = x_t, K_{t-1}, a_t) = P(X_{t+1} = x_{t+1} | X_t = x_t, a_t)$$

- Furthermore, the action a_t does not affect the state X_t , and therefore

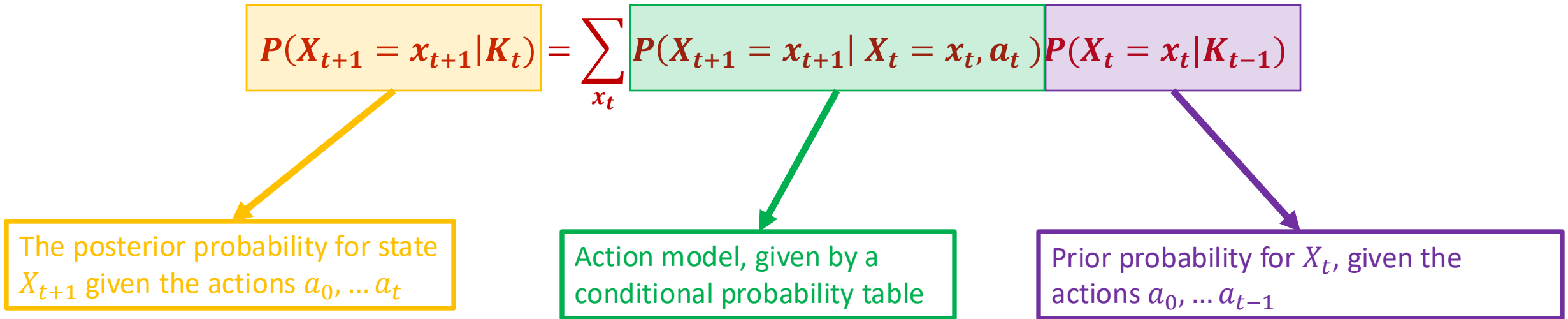
$$P(X_t = x_t | K_{t-1}, a_t) = P(X_t = x_t | K_{t-1})$$

Substitute these into the equation above, and we obtain:

$$P(X_{t+1} = x_{t+1} | K_t) = \sum_{x_t} P(X_{t+1} = x_{t+1} | X_t = x_t, a_t) P(X_t = x_t | K_{t-1})$$

Posterior probabilities

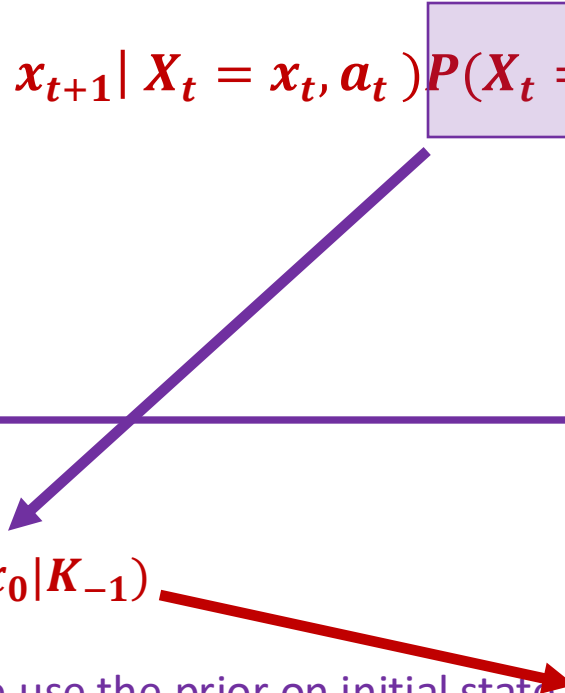
- Let's take a closer look at this result:



How do we know $P(X_t = x_t | K_{t-1})$?

Posterior probabilities

- Let's take a closer look at this result:

$$P(X_{t+1} = x_{t+1} | K_t) = \sum_{x_t} P(X_{t+1} = x_{t+1} | X_t = x_t, a_t) P(X_t = x_t | K_{t-1})$$


For $t = 0$, the prior takes the form:

$$P(X_0 = x_0 | K_{-1})$$

But we never have $t = -1$, so for the base case at $t = 0$, we use the prior on initial state, $P(X_0 = x_0)$, which gives:

$$P(X_1 = x_1 | K_0) = \sum_{x_0} P(X_1 = x_1 | X_0 = x_0, a_0) P(X_0 = x_0)$$

We now proceed iteratively to compute $P(X_{t+1} = x_{t+1} | K_t)$ for arbitrary t .

Posterior probabilities

- Let's take a closer look at this result:

$$P(X_{t+1} = x_{t+1} | K_t) = \sum_{x_t} P(X_{t+1} = x_{t+1} | X_t = x_t, a_t) P(X_t = x_t | K_{t-1})$$



The sum is taken over the set of all possible values for x_t

$$\sum_{x_t \in \{L, K, O, H, D\}} P(X_{t+1} = x_{t+1} | X_t = x_t, a_t) P(X_t = x_t | K_{t-1})$$

At time t , the state could be any of the rooms, $\{L, K, O, H, D\}$.

Posterior probabilities

- Let's take a closer look at this result:

$$P(X_{t+1} = \boxed{x_{t+1}} | K_t) = \sum_{x_t} P(X_{t+1} = \boxed{x_{t+1}} | X_t = x_t, \boxed{a_t}) P(X_t = x_t | K_{t-1})$$

Don't forget, a_t is hiding in K_t .

This equation applies to a specific action, a_t , e.g., move up.

If we want to know the probability distribution of X_{t+1} for a different action, e.g., move right, we need to use the equation again.

This equation tells us how to compute the probability that X_{t+1} is in the specific state, $x_{t+1} \in \{L, K, O, H, D\}$.

To fully understand where we are we need to use this equation five times, once for each possible value for X_{t+1} .

Belief state

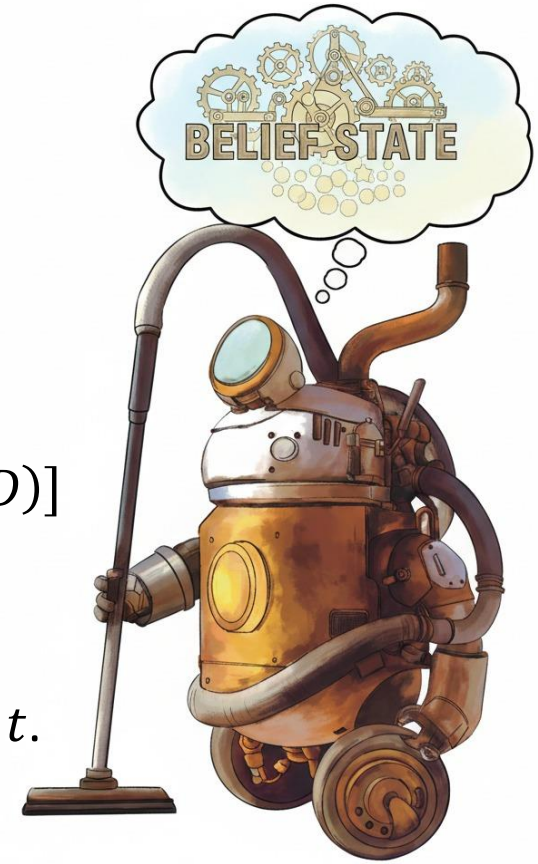
- It will sometimes be convenient to refer to the entire probability distribution at time t .
- We refer to this distribution as the belief state at time t , denoted by b_t .
- The **belief state** is a row vector whose elements correspond to the possible states.
- In our case, there are five possible states, so $b_t^T \in \mathbb{R}^5$.
- At $t = 0$, the belief state is merely our initial distribution:

$$\begin{aligned} b_0 &= [P(X_0 = L), \quad P(X_0 = K) \quad P(X_0 = O) \quad P(X_0 = H) \quad P(X_0 = D)] \\ &= [0 \quad 0 \quad 1 \quad 0 \quad 0] \end{aligned}$$

- The belief state b_{t+1} is conditioned on the initial state x_0 and all actions taken until time t .

$$b_{t+1}^T = \begin{bmatrix} P(X_{t+1} = L \mid a_1 \dots a_t, x_0) \\ P(X_{t+1} = K \mid a_1 \dots a_t, x_0) \\ P(X_{t+1} = O \mid a_1 \dots a_t, x_0) \\ P(X_{t+1} = H \mid a_1 \dots a_t, x_0) \\ P(X_{t+1} = D \mid a_1 \dots a_t, x_0) \end{bmatrix}$$

➤ Note that we use b_{t+1}^T to denote the transpose of b_{t+1} (for formatting purposes).



Matrix form

- We can write the expression for $P(X_{t+1} = x_{t+1} | K_t)$ in a more compact form
- To keep things simple, let's use the action, *move right*, and compute the probability of arriving to the *Living Room*:

$$\begin{aligned}
 P(X_{t+1} = L | K_t) &= \sum_{x_t} P(X_{t+1} = L | X_t = x_t, a_t = r) P(X_t = x_t | K_{t-1}) \\
 &= P(X_{t+1} = L | X_t = L, r) P(X_t = L | K_{t-1}) + \\
 &\quad P(X_{t+1} = L | X_t = K, r) P(X_t = K | K_{t-1}) + \\
 &\quad P(X_{t+1} = L | X_t = O, r) P(X_t = O | K_{t-1}) + \\
 &\quad P(X_{t+1} = L | X_t = H, r) P(X_t = H | K_{t-1}) + \\
 &\quad P(X_{t+1} = L | X_t = D, r) P(X_t = D | K_{t-1})
 \end{aligned}$$

- We can write this as a simple matrix equation:

$$P(X_{t+1} = L | K_t) = [P(X_t = L | K_{t-1}) \quad P(X_t = K | K_{t-1}) \quad P(X_t = O | K_{t-1}) \quad P(X_t = H | K_{t-1}) \quad P(X_t = D | K_{t-1})] \begin{bmatrix} P(L|L, r) \\ P(L|K, r) \\ P(L|O, r) \\ P(L|H, r) \\ P(L|D, r) \end{bmatrix}$$

Matrix form

- We can write the expression for $P(X_{t+1} = x_{t+1} | K_t)$ in a more compact form
- To keep things simple, let's use the action, *move right*, and compute the probability of arriving to the *Living Room*:

$$\begin{aligned}
 P(X_{t+1} = L | K_t) &= \sum_{x_t} P(X_{t+1} = L | X_t = x_t, a_t = r) P(X_t = x_t | K_{t-1}) \\
 &= P(X_{t+1} = L | X_t = L, r) P(X_t = L | K_{t-1}) + \\
 &\quad P(X_{t+1} = L | X_t = K, r) P(X_t = K | K_{t-1}) + \\
 &\quad P(X_{t+1} = L | X_t = O, r) P(X_t = O | K_{t-1}) + \\
 &\quad P(X_{t+1} = L | X_t = H, r) P(X_t = H | K_{t-1}) + \\
 &\quad P(X_{t+1} = L | X_t = D, r) P(X_t = D | K_{t-1})
 \end{aligned}$$

This is merely one column from the conditional probability matrix for the action *move right*.

- We can write this as a simple matrix equation:

This row matrix is exactly the prior b_t

$$P(X_{t+1} = L | K_t) = [P(X_t = L | K_{t-1}) \quad P(X_t = K | K_{t-1}) \quad P(X_t = O | K_{t-1}) \quad P(X_t = H | K_{t-1}) \quad P(X_t = D | K_{t-1})]$$

$$\begin{bmatrix} P(L|L,r) \\ P(L|K,r) \\ P(L|O,r) \\ P(L|H,r) \\ P(L|D,r) \end{bmatrix}$$

Matrix form

- We can write a similar expression for each state $X_{t+1} \in \{L, K, O, H, D\}$.
- We can then collect these five equations into a single matrix equation.
- Let $M_{\mathcal{A}}$ denote the conditional probability matrix for action $\mathcal{A}$.
- Recall, when $\mathcal{A}$ is *move right*, we have:

$$M_r = \begin{bmatrix} P(L|L,r) & P(K|L,r) & P(O|L,r) & P(H|L,r) & P(D|L,r) \\ P(L|K,r) & P(K|K,r) & P(O|K,r) & P(H|K,r) & P(D|K,r) \\ P(L|O,r) & P(K|O,r) & P(O|O,r) & P(H|O,r) & P(D|O,r) \\ P(L|H,r) & P(K|H,r) & P(O|H,r) & P(H|H,r) & P(D|H,r) \\ P(L|D,r) & P(K|D,r) & P(O|D,r) & P(H|D,r) & P(D|D,r) \end{bmatrix} = \begin{bmatrix} 0.2 & 0.8 & 0.0 & 0.0 & 0.0 \\ 0.0 & 1.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.2 & 0.8 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.2 & 0.8 \\ 0.0 & 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix}$$

- We can now compute $b_{t+1} = P(X_{t+1} = L|K_t)$ by combining these equations to obtain:

$$b_{t+1} = b_t M_r$$

Let's take a closer look....

Conditional probability tables

We can construct a conditional probability table for each action using this large table.

<i>X1</i>	<i>A1</i>	Living Room	Kitchen	Office	Hallway	Dining Room
Living Room	L	1	0	0	0	0
Living Room	R	0.2	0.8	0	0	0
Living Room	U	1	0	0	0	0
Living Room	D	0.2	0	0	0.8	0
Kitchen	L	0.8	0.2	0	0	0
Kitchen	R	0	1	0	0	0
Kitchen	U	0	1	0	0	0
Kitchen	D	0.2	0	0	0	0.8
Office	L	0	0	1	0	0
Office	R	0	0	0.2	0.8	0
Office	U	0	0	1	0	0
Office	D	0	0	1	0	0
Hallway	L	0	0	0.8	0.2	0
Hallway	R	0	0	0	0.2	0.8
Hallway	U	0.8	0	0	0.2	0
Hallway	D	0	0	0	1	0
Dining Room	L	0	0	0	0.8	0.2
Dining Room	R	0	0	0	0	1
Dining Room	U	0	0.8	0	0	0.2
Dining Room	D	0	0	0	0	1

- Consider the action move right.
- We construct the conditional probability matrix for this action by collecting the move right rows from the table.

$$M_r = \begin{bmatrix} 0.2 & 0.8 & 0.0 & 0.0 & 0.0 \\ 0.0 & 1.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.2 & 0.8 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.2 & 0.8 \\ 0.0 & 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix}$$

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Kitchen	D	0.2	0	0	0	0.8
Office	L	0	0	1	0	0
Office	R	0	0	0.2	0.8	0
Office	U	0	0	1	0	0
Office	D	0	0	1	0	0
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Hallway	R	0	0	0	0.2	0.8
Hallway	U	0.8	0	0	0.2	0
Hallway	D	0	0	0	1	0
Dining Room	L	0	0	0	0.8	0.2
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Kitchen	D	0.2	0	0	0	0.8
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$$M_r = \begin{bmatrix} P(L|L,r) & P(K|L,r) & P(O|L,r) & P(H|L,r) & P(D|L,r) \\ P(L|K,r) & P(K|K,r) & P(O|K,r) & P(H|K,r) & P(D|K,r) \\ P(L|O,r) & P(K|O,r) & P(O|O,r) & P(H|O,r) & P(D|O,r) \\ P(L|H,r) & P(K|H,r) & P(O|H,r) & P(H|H,r) & P(D|H,r) \\ P(L|D,r) & P(K|D,r) & P(O|D,r) & P(H|D,r) & P(D|D,r) \end{bmatrix}$$

Calculating the belief state

$$b_{t+1} = b_t M_r =$$

$$\begin{bmatrix} P(X_t = L | K_{t-1}) & P(X_t = K | K_{t-1}) & P(X_t = O | K_{t-1}) & P(X_t = H | K_{t-1}) & P(X_t = D | K_{t-1}) \end{bmatrix} \begin{bmatrix} P(L|L,r) & P(K|L,r) & P(O|L,r) & P(H|L,r) & P(D|L,r) \\ P(L|K,r) & P(K|K,r) & P(O|K,r) & P(H|K,r) & P(D|K,r) \\ P(L|O,r) & P(K|O,r) & P(O|O,r) & P(H|O,r) & P(D|O,r) \\ P(L|H,r) & P(K|H,r) & P(O|H,r) & P(H|H,r) & P(D|H,r) \\ P(L|D,r) & P(K|D,r) & P(O|D,r) & P(H|D,r) & P(D|D,r) \end{bmatrix}$$

Let's look at the first entry of b_{t+1} --- the product of b_t and the first column of M_r .

Calculating the belief state

$$b_{t+1} = b_t M_r =$$

$$\begin{bmatrix} P(X_t = L | K_{t-1}) & P(X_t = K | K_{t-1}) & P(X_t = O | K_{t-1}) & P(X_t = H | K_{t-1}) & P(X_t = D | K_{t-1}) \end{bmatrix} \begin{bmatrix} P(L|L,r) & P(K|L,r) & P(O|L,r) & P(H|L,r) & P(D|L,r) \\ P(L|K,r) & P(K|K,r) & P(O|K,r) & P(H|K,r) & P(D|K,r) \\ P(L|O,r) & P(K|O,r) & P(O|O,r) & P(H|O,r) & P(D|O,r) \\ P(L|H,r) & P(K|H,r) & P(O|H,r) & P(H|H,r) & P(D|H,r) \\ P(L|D,r) & P(K|D,r) & P(O|D,r) & P(H|D,r) & P(D|D,r) \end{bmatrix}$$

$$\begin{aligned} \rightarrow & P(X_{t+1} = L | X_t = L, r) P(X_t = L | K_{t-1}) + \\ & P(X_{t+1} = L | X_t = K, r) P(X_t = K | K_{t-1}) + \\ & P(X_{t+1} = L | X_t = O, r) P(X_t = O | K_{t-1}) + \\ & P(X_{t+1} = L | X_t = H, r) P(X_t = H | K_{t-1}) + \\ & P(X_{t+1} = L | X_t = D, r) P(X_t = D | K_{t-1}) \end{aligned}$$

- ***This is exactly the computation we performed above!***
- ***This works for each entry of b_{t+1} .***

Calculating the belief state

If we execute action $\mathcal{A}$ at time t , the belief state, $b_{t+1} = P(X_{t+1} = x_{t+1} \mid K_t)$, is given by

$$b_{t+1} = b_t M_{\mathcal{A}}$$

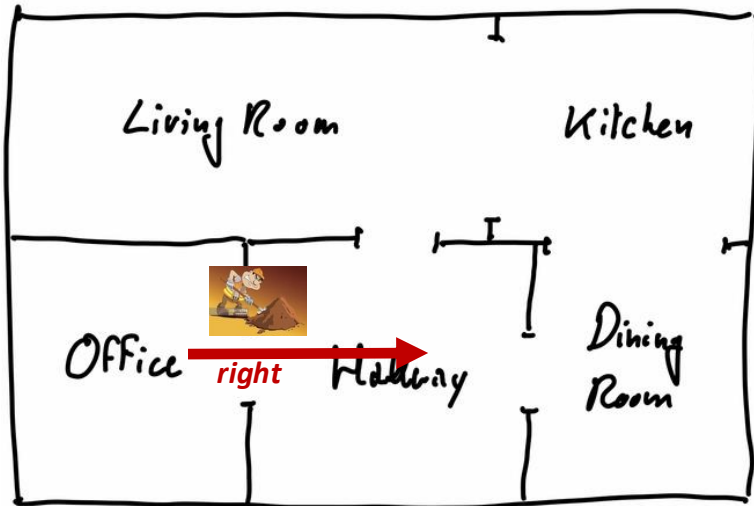
in which $M_{\mathcal{A}}$ is the conditional probability matrix for action $\mathcal{A}$ and b_t is the belief state at time t .

Example: Move Right

As we have seen above, if we execute the command move right from the initial state, $x_0 = Office$, we obtain

$$b_1 = b_0 M_r = [0 \ 0 \ 1 \ 0 \ 0] \begin{bmatrix} 0.2 & 0.80 & 0.0 & 0.0 & 0.0 \\ 0.0 & 1.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.2 & 0.8 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.2 & 0.8 \\ 0.0 & 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix} = [0.0 \quad 0.0 \quad 0.2 \quad 0.8 \quad 0.0]$$

$$= [P(X_1 = L \mid r, X_0 = O), \quad P(X_1 = K \mid r, X_0 = O) \quad P(X_1 = O \mid r, X_0 = O), \quad P(X_1 = H \mid r, X_0 = O) \quad P(X_1 = D \mid r, X_0 = O)]$$

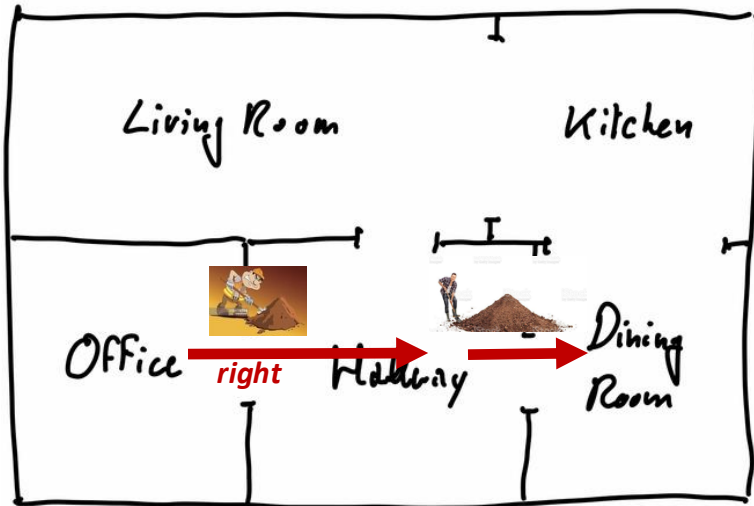


- You can imagine probability mass being pushed to the right by a sloppy worker.
- Only 80% of the probability arrives to the Hallway.

Move right multiple times

If we now again execute the action move right at time $t = 1$, we obtain

$$b_2 = b_1 M_r = [0.0 \quad 0.0 \quad 0.2 \quad 0.8 \quad 0.0] \begin{bmatrix} 0.2 & 0.80 & 0.0 & 0.0 & 0.0 \\ 0.0 & 1.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.2 & 0.8 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.2 & 0.8 \\ 0.0 & 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix} = [0.0 \quad 0.0 \quad 0.04 \quad 0.32 \quad 0.64]$$



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If we execute the action *move right* n times in succession, we obtain

$$b_n = b_0 M_r^n$$

➤ *As you can imagine, there's a very beautiful theory to systems like this – a combination of linear algebra and probability theory.*

Markov chains

- Suppose we have chosen a specific sequence of actions: $a_0, \dots, a_n$
- At stage $t + 1$, we compute the belief b_{t+1} using conditional probability matrix M_{a_t} and the prior belief b_t :

$$b_{t+1} = b_t M_{a_t} = \underbrace{b_0 M_{a_0}}_{b_1} \underbrace{M_{a_1} M_{a_2} \dots M_{a_{t-1}}}_{b_t} M_{a_t}$$

The diagram illustrates the recursive calculation of belief states. It shows the equation $b_{t+1} = b_t M_{a_t} = b_0 M_{a_0} M_{a_1} M_{a_2} \dots M_{a_{t-1}} M_{a_t}$. Brackets are used to group terms and show how each belief state is derived from the previous one and the next action:

- A blue bracket under $b_0 M_{a_0}$ is labeled b_1 .
- A green bracket under $M_{a_1} M_{a_2}$ is labeled b_2 .
- A brown bracket under $M_{a_2} \dots M_{a_{t-1}}$ is labeled b_3 .
- A purple bracket under $M_{a_{t-1}} M_{a_t}$ is labeled b_t .

Vertical dotted lines connect the end of each bracket to the start of the next one, showing the sequence of calculations. Three vertical dots between the brown and purple brackets indicate the continuation of the sequence.

Recall that b_0 is the initial distribution for state, in our example scenario:

$$b_0 = [0 \ 0 \ 1 \ 0 \ 0]$$

At any time t , all of the available information about the history of the robot (where it has been, what it has done) is contained in the belief state b_t .

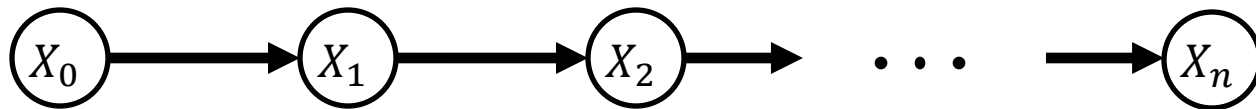
If you know b_t , learning specific previous actions does not add information.

Markov chains

- A sequence of random variables $X_0, \dots, X_n$ is a Markov chain if the Markov property holds

$$P(X_{t+1} | X_t, X_{t-1}, \dots, X_0) = P(X_{t+1} | X_t)$$

- In our case, we have a fixed action sequence $a_0 \dots a_t$, which defines the distributions for each of the X_i .
- For a fixed sequence of actions, the state of our vacuum cleaning robot forms a Markov chain.
- A Markov chain has a simple graphical representation:

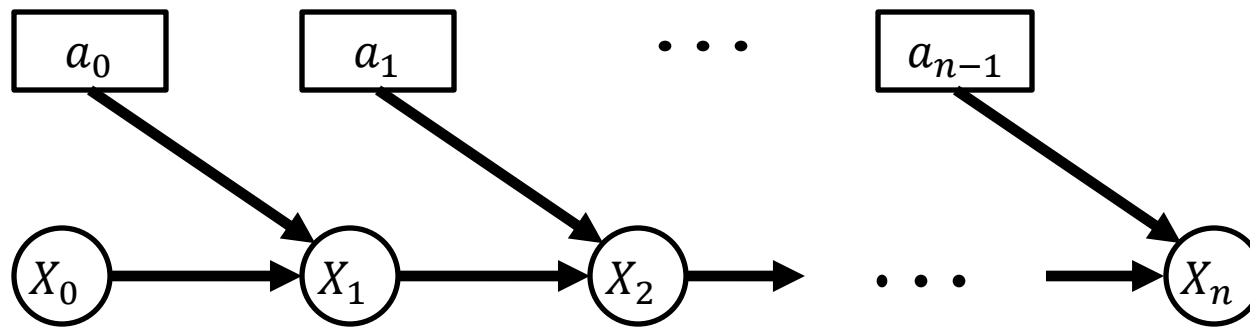


$$P(X_{t+1} | a_0 \dots a_t, x_0) \rightarrow b_{t+1} = b_t M_{a_t}$$

Each node includes the distribution b_t and each arc corresponds to the computation $b_{t-1} M_{a_t}$

Controlled Markov chains

- So far, in our discussions about the Markov chain $X_0, \dots, X_t$, we have been careful to always add the phrase *“for a fixed action sequence $a_0 \dots a_t$.”*
- We can think of the actions, $a_0 \dots a_t$, as **control inputs** to the system.
- Our choice of $a_0 \dots a_t$ controls how the system evolves.
- We don't control the actual state X_t , but we do control which conditional probability matrix is used to update the belief state.
- We call this kind of process a **controlled Markov chain**.
- A controlled Markov chain also has a nice graphical representation:



Note:

States are *random* – circles.

Actions are *deterministic* – boxes.

There is so much important stuff on this slide!!

Next Lecture: A Vacuum Cleaning Robot

- Bayes Nets
- Uncertainty in sensing for a sequence of measurements
- Hidden Markov Models (HMM)